

W4

Threat Detection Platform

4
DAYS

Project Information

Project	Threat Detection Platform
Week	Week 4 — Dashboard Visualization
Dashboard URL	http://127.0.0.1:5000
Data Source	Python + MongoDB + Flask
Stack	Python + Flask + MongoDB + HTML

Week 4 Objectives

- By the end of Week 4, the Threat Detection Platform should be able to:
 - create a web-based dashboard
 - display threat intelligence records
 - visualize alerts from MongoDB
 - monitor HIGH and MEDIUM risk threats
 - build a SOC-style monitoring interface
 - integrate Flask with MongoDB

DAY 15

Flask Dashboard Setup

STEP 1 — Install Flask (Goal: Setup web dashboard environment)

```
pip install flask
```

Expected Output

Successfully installed flask

STEP 2 — Create app.py

(Goal: Connect Flask with MongoDB)

```
from flask import Flask, render_template
from db import collection, db

app = Flask(__name__)

@app.route("/")
def dashboard():

    threats = list(collection.find({}, {"_id": 0}))

    alerts = list(db["alerts"].find({}, {"_id": 0}))

    return render_template(
        "dashboard.html",
        threats=threats,
        alerts=alerts
    )

if __name__ == "__main__":
    app.run(debug=True)
```

Expected Output

Flask application created successfully

STEP 3 — Create dashboard.html

(Goal: Build SOC-style monitoring dashboard)

```
<!DOCTYPE html>
<html>
<head>
    <title>Threat Detection Dashboard</title>
</head>

<body>

<h1>Threat Detection Dashboard</h1>

<h2>Threat Intelligence</h2>

<table border="1">

<tr>
```

```

        <th>IP Address</th>
        <th>Malicious Score</th>
        <th>Risk Level</th>
    </tr>

    {% for threat in threats %}
    <tr>
        <td>{{ threat.ip }}</td>
        <td>{{ threat.malicious_score }}</td>
        <td>{{ threat.risk_level }}</td>
    </tr>
    {% endfor %}

</table>

<h2>Alerts</h2>

<table border="1">

<tr>
    <th>IP Address</th>
    <th>Risk Level</th>
    <th>Message</th>
</tr>

{% for alert in alerts %}
<tr>
    <td>{{ alert.ip }}</td>
    <td>{{ alert.risk_level }}</td>
    <td>{{ alert.message }}</td>
</tr>
{% endfor %}

</table>

</body>
</html>

```

Expected Output

Dashboard template ready

DAY 19

Run Dashboard Application

STEP 4 — Execute Flask Dashboard (Goal: Launch dashboard visualization)

```
python app.py
```

Expected Output

Running on http://127.0.0.1:5000

Completion Checklist

- ✓ Flask installed successfully
- ✓ Dashboard application created
- ✓ MongoDB connected with Flask
- ✓ Threat intelligence visualization completed
- ✓ Alert monitoring dashboard created
- ✓ SOC-style dashboard interface working
- ✓ Week 4 pipeline executed successfully

Week 4 — Required GitHub Commits

```
git commit -m "feat: add Flask dashboard application"
git commit -m "feat: add dashboard visualization template"
git commit -m "feat: integrate MongoDB with dashboard"
git commit -m "test: verify week 4 dashboard pipeline"
```

PROJECT COMPLETE!
